PAPER_792: Large Magellanic Cloud — Three-UQFF Irregular Satellite Galaxy

Author: Daniel T. Murphy
Framework: UQFF (Universal Quantum Field Superconductive Framework) — Three-UQFF Simultaneous
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Abstract

The Large Magellanic Cloud (LMC) is the largest satellite galaxy of the Milky Way, located ~163,000 light-years away ($z \approx 0.0005$) in the southern constellations Dorado and Mensa. With ~10^{10} M⊙ total mass and a diameter of ~14,000 ly, the LMC is an irregular (Irr) galaxy showing a disrupted barred spiral structure. It hosts the most actively star-forming region in the Local Group — the Tarantula Nebula (30 Doradus, PAPER_774). Three-UQFF simultaneous analysis of the LMC as a whole yields g_primary ≈ 1.053×10^{-3} m/s^2 at the galaxy-wide scale ($v = 10^5$ m/s standard disk rotation).

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1. Introduction

The LMC's proximity makes it a unique laboratory: individual stars, clusters (NGC 1805, PAPER_787), and the Tarantula hyperstar-forming region (PAPER_774) are all resolvable. As a whole, the LMC has a rotation velocity of ~50–80 km/s — somewhat slower than standard spirals, reflecting its lower mass. However, the Tarantula Nebula starburst region within it operates at the 10^6 m/s starburst regime. The Three-UQFF whole-LMC analysis uses galaxy-scale parameters ($M = 10^{10}$ M⊙, $r = 6.62\times10^{19}$ m, $v = 10^5$ m/s) rather than the starburst sub-region parameters of PAPER_774.

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2. Three-UQFF Parameters

Parameter	Symbol	Value	Source
LMC total mass	M	10^{10} M⊙ = 1.989×10^{40} kg	HI surveys
LMC radius	r	6.62×10^{19} m (~7 kly)	Angular size
SFR (LMC-wide)	—	0.5 M⊙/yr	Average
Age	t	5×10^9 yr = 1.578×10^{17} s	Hubble time
M_sf	—	0.05	UQFF LMC SFR
Redshift	z	0.0005	Distance
v_EM	v	10^5 m/s	LMC rotation
B_EM	B	10^{-5} T	LMC field

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3. Three-UQFF Derivation

Mode 1: Compressed UQFF

g_grav = 6.6743e-11 × 1.989e40 / (6.62e19)^2

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    = 1.328e30 / 4.382e39 = 3.030e-10 m/s^2

H(z)×t = 2.267e-18 × 1.578e17 = 0.358; factor = 1.358
factor_sf = 1.05; factor_TRZ = 1.04
g_grav_total = 3.030e-10 × 1.358 × 1.05 × 1.04 = 4.499e-10 m/s^2
a_EM = 1.053e-3 m/s^2
g_comp = 1.053e-3 m/s^2
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Mode 2: Resonant UQFF

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g_res = 1.053e-3 × 1.000285 = 1.053e-3 m/s^2
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Mode 3: Buoyancy UQFF

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V = (4/3)π(6.62e19)^3 = 1.217e60 m^3; a_Ubi << a_EM
g_buoy = 1.053e-3 m/s^2
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Three-UQFF Simultaneous Result

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g_compressed = 1.053e-3 m/s^2
g_resonant   = 1.053e-3 m/s^2
g_buoyancy   = 1.053e-3 m/s^2
g_primary = 1.053e-3 m/s^2

Note: Compare PAPER_774 (Tarantula 30 Dor within LMC): g = 1.053e-1 m/s^2 at starburst v=1e6, B=1e-4
      LMC whole-galaxy result uses galaxy-scale parameters, not subregion.
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4. Physical Interpretation

The LMC's galaxy-wide properties at v = 10^5 m/s yield standard UQFF result g = 1.053×10^{-3} m/s^2. This coexists with the Tarantula starburst subregion (PAPER_774, g = 1.053×10^{-1} m/s^2) because UQFF scales with local EM parameters. This demonstrates UQFF multi-scale consistency: the galaxy-wide average uses global rotation velocity, while local starburst regions use their own extreme local velocities. The LMC's Milky Way tidal interaction, which shapes its irregular morphology, does not alter UQFF's electromagnetic Aether coupling at the global galaxy scale.

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5. Conclusions

Three-UQFF applied to the LMC yields g_primary $\approx 1.053\times10^{-3}$ m/s^2 galaxy-wide. UQFF multi-scale consistency confirmed: the LMC whole-galaxy result coexists with the Tarantula starburst result (PAPER_774, g = 1.053×10^{-1} m/s^2) by using scale-appropriate v and B parameters.

PAPER_792, CP4 Three-UQFF class #376. v5.42.